Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 6 Solutions

Problem 6.1

You have been hired to investigate a disturbing connection between ice cream sales and crime. You are given a report that includes the joint distribution of random variable $S \in \{\texttt{low}, \texttt{high}\}$ that represents ice cream sales and random variable $X \in \{\texttt{low}, \texttt{high}\}$ that represents the crime rate in the city.

s	x	$p_{S,X}(s,x)$
low	low	0.15
low	high	0.15
high	low	0.3
high	high	0.4

(a) Are random variables S and X independent?

Solution: No. We marginalize the joint distribution $p_{S,X}$ above to obtain the marginal distributions

$$\begin{aligned} p_S(\texttt{low}) &= p_{S,X}(\texttt{low},\texttt{low}) + p_{S,X}(\texttt{low},\texttt{high}) = 0.15 + 0.15 = 0.3, \\ p_S(\texttt{high}) &= 1 - p_S(\texttt{low}) = 0.7, \\ p_X(\texttt{low}) &= p_{S,X}(\texttt{low},\texttt{low}) + p_{S,X}(\texttt{high},\texttt{low}) = 0.15 + 0.3 = 0.45, \\ p_X(\texttt{high}) &= 1 - p_X(\texttt{low}) = 0.55. \end{aligned}$$

To show that the two random variables are not independent it suffices to show that

$$p_{S,X}(s,x) \neq p_S(s)\, p_X(x)$$

for some pair (s,x). Indeed,

$$p_S(\texttt{low})\, p_X(\texttt{low}) = 0.135 \neq 0.15 = p_{S,X}(\texttt{low},\texttt{low})$$

and we conclude that S and X are not independent.

(b) Assume that the trends of ice cream sales and crime during any day are independent of all other days. Suppose the ice cream sales were low on Monday and high on Tuesday. On which day is the crime rate more likely to be high?

Solution: We compute the conditional probability of high crime volume,

$$\begin{aligned} p_{X|S}(\texttt{high}|\texttt{low}) &= \frac{p_{S,X}(\texttt{low},\texttt{high})}{p_S(\texttt{low})} = \frac{0.15}{0.3} = 0.5, \\ p_{X|S}(\texttt{high}|\texttt{high}) &= \frac{p_{S,X}(\texttt{high},\texttt{high})}{p_S(\texttt{high})} = \frac{0.4}{0.7} \approx 0.57, \end{aligned}$$

and conclude that crime is more likely to be high on Tuesday. This seems to suggest that increase in ice cream sales leads to increase in crime rates!

(c) After further investigation, you obtain additional information about the dependencies of the ice cream sales and crime volume on the temperature on that day. Let $T \in \{\texttt{low}, \texttt{high}\}$ denote the temperature, and assume that the conditional distributions of S and X given T are as follows:

s	x	$p_{S,X\|T}(s, x\|\texttt{low})$
low	low	0.36
low	high	0.24
high	low	0.24
high	high	0.16

s	x	$p_{S,X\|T}(s, x\|\texttt{high})$
low	low	0.08
low	high	0.12
high	low	0.32
high	high	0.48

Are random variables S and X conditionally independent given T?

Solution: Yes. We marginalize the conditional joint distribution of S and X given $T = \texttt{low}$ to obtain

$$\begin{aligned}
p_{S|T}(\texttt{low}|\texttt{low}) &= p_{S,X|T}(\texttt{low}, \texttt{low}|\texttt{low}) + p_{S,X|T}(\texttt{low}, \texttt{high}|\texttt{low}) = 0.36 + 0.24 = 0.6 \\
p_{S|T}(\texttt{high}|\texttt{low}) &= 1 - p_{S|T}(\texttt{low}|\texttt{low}) = 0.4, \\
p_{X|T}(\texttt{low}|\texttt{low}) &= p_{S,X|T}(\texttt{low}, \texttt{low}|\texttt{low}) + p_{S,X|T}(\texttt{high}, \texttt{low}|\texttt{low}) = 0.36 + 0.24 = 0.6, \\
p_{X|T}(\texttt{high}|\texttt{low}) &= 1 - p_{X|T}(\texttt{low}|\texttt{low}) = 0.4,
\end{aligned}$$

and proceed to compute

$$p_{S|T}(s|\texttt{low})\, p_{X|T}(x|\texttt{low}) = \begin{cases} 0.6 \cdot 0.6 = 0.36 & s = \texttt{low},\ x = \texttt{low} \\ 0.6 \cdot 0.4 = 0.24 & s = \texttt{low},\ x = \texttt{high} \\ 0.4 \cdot 0.6 = 0.24 & s = \texttt{high},\ x = \texttt{low} \\ 0.4 \cdot 0.4 = 0.16 & s = \texttt{high},\ x = \texttt{high} \end{cases}$$

It is now straightforward to verify that $p_{S,X|T}(s, x|\texttt{low}) = p_{S|T}(s|\texttt{low})\, p_{X|T}(x|\texttt{low})$ for all $s, x \in \{\texttt{low}, \texttt{high}\}$.

Similarly, for $T = \texttt{high}$ we have

$$\begin{aligned}
p_{S|T}(\texttt{low}|\texttt{high}) &= p_{S,X|T}(\texttt{low}, \texttt{low}|\texttt{high}) + p_{S,X|T}(\texttt{low}, \texttt{high}|\texttt{high}) = 0.08 + 0.12 = 0.2, \\
p_{S|T}(\texttt{high}|\texttt{high}) &= 1 - p_{S|T}(\texttt{low}|\texttt{high}) = 0.8, \\
p_{X|T}(\texttt{low}|\texttt{high}) &= p_{S,X|T}(\texttt{low}, \texttt{low}|\texttt{high}) + p_{S,X|T}(\texttt{high}, \texttt{low}|\texttt{high}) = 0.08 + 0.32 = 0.4, \\
p_{X|T}(\texttt{high}|\texttt{high}) &= 1 - p_{X|T}(\texttt{low}|\texttt{high}) = 0.6.
\end{aligned}$$

We compute

$$p_{S|T}(s|\texttt{high})\, p_{X|T}(x|\texttt{high}) = \begin{cases} 0.2 \cdot 0.4 = 0.08 & s = \texttt{low},\ x = \texttt{low} \\ 0.2 \cdot 0.6 = 0.12 & s = \texttt{low},\ x = \texttt{high} \\ 0.8 \cdot 0.4 = 0.32 & s = \texttt{high},\ x = \texttt{low} \\ 0.8 \cdot 0.6 = 0.48 & s = \texttt{high},\ x = \texttt{high}. \end{cases}$$

and verify that $p_{S,X|T}(s,x|\texttt{high}) = p_{S|T}(s|\texttt{high})\,p_{X|T}(x|\texttt{high})$ for all $s,x \in \{\texttt{low},\texttt{high}\}$.

We therefore have established that $S \perp\!\!\!\perp X|T$ and that random variables S, T, and X form a Markov chain $S \leftrightarrow T \leftrightarrow X$. The additional factor of the temperature explains the changes in ice cream sales and crime rates and removes the dependency between them that was observed in part (b) when temperature was not considered.

(d) Sort the quantities below in the increasing order:

$$I(S;X|T) \qquad I(X;S) \qquad I(S;X|T=\texttt{high}).$$

Solution: Since mutual information is symmetric and we established in part (a) that random variables S and X are dependent, we have $I(X;S) = I(S;X) > 0$.

Since we established in part (c) that $S \perp\!\!\!\perp X|\,(T=\texttt{high})$ and $S \perp\!\!\!\perp X|T$, we have $I(S;X|\,(T=\texttt{high})) = I(S;X|T) = 0$.

Combining, we obtain

$$0 = I(S;X|\,(T=\texttt{high})) = I(S;X|T) < I(X;S).$$

(e) Can you determine the distribution $p_T(\cdot)$ from the tables above? If so, find it. If not, explain what additional information is needed to determine $p_T(\cdot)$.

Solution: Yes. The Total Probability Law implies that for any $s,x \in \{\texttt{low},\texttt{high}\}$,

$$\begin{aligned} p_{S,X}(s,x) &= p_T(\texttt{low})\,p_{S,X|T}(s,x|\texttt{low}) + p_T(\texttt{high})\,p_{S,X|T}(s,x|\texttt{high}) \\ &= p_T(\texttt{low})\,p_{S,X|T}(s,x|\texttt{low}) + (1-p_T(\texttt{low}))\,p_{S,X|T}(s,x|\texttt{high}), \end{aligned}$$

and thus

$$p_T(\texttt{low}) = \frac{p_{S,X}(s,x) - p_{S,X|T}(s,x|\texttt{high})}{p_{S,X|T}(s,x|\texttt{low}) - p_{S,X|T}(s,x|\texttt{high})}.$$

By substituting the values of the appropriate PMFs for $s=\texttt{low}$ and $x=\texttt{low}$, we obtain

$$p_T(\texttt{low}) = \frac{0.15-0.08}{0.36-0.08} = \frac{1}{4}.$$

We leave it as an exercise to verify that substituting other pairs of values for s and x yields the same answer, i.e., the tables in part (a) and part (c) are consistent with

$$p_T(t) = \begin{cases} 1/4 & t=\texttt{low} \\ 3/4 & t=\texttt{high}. \end{cases}$$